Inter-VLAN Routing Basics

Cisco Packet Tracer Lab Report

1. Theory: Methods of Inter-VLAN Routing

Devices in different VLANs are in different broadcast domains and different IP subnets, so they need a Layer 3 device to communicate.

- Router-on-a-stick: one physical router interface is connected to a switch trunk port and split into logical sub-interfaces, one per VLAN. Each sub-interface is the default gateway for its VLAN.
- Layer 3 switch: the switch routes between VLANs itself using SVIs (Switch Virtual Interfaces, for example interface vlan 10) with ip routing enabled.
- Legacy method: one router interface per VLAN. It does not scale and is rarely used.

2. Task 1: Topology and VLANs

Devices: 1 router (2911), 1 switch (2960), 3 PCs. Switch Fa0/1 connects to router Gi0/0 and is a trunk.

VLAN	Name (Service)	Switch Port	PC IP Address	Gateway
10	Foodies (Zomato)	Fa0/2 (PC1)	192.168.10.10/24	192.168.10.1
20	Shoppers (Flipkart)	Fa0/3 (PC2)	192.168.20.10/24	192.168.20.1
30	Streamers (Spotify)	Fa0/4 (PC3)	192.168.30.10/24	192.168.30.1

Switch configuration:

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name Foodies
Switch(config-vlan)# vlan 20
Switch(config-vlan)# name Shoppers
Switch(config-vlan)# vlan 30
Switch(config-vlan)# name Streamers
Switch(config-vlan)# exit
Switch(config)# interface fastEthernet 0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# interface fastEthernet 0/3
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# interface fastEthernet 0/4
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 30
Switch(config-if)# interface fastEthernet 0/1
Switch(config-if)# switchport mode trunk
Switch(config-if)# end
Switch# copy running-config startup-config
```

3. Task 2: Router-on-a-Stick Configuration

The physical interface has no IP address and must be turned on. Each sub-interface needs the VLAN tag and the gateway IP:

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface gigabitEthernet 0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)# exit
Router(config)# interface gigabitEthernet 0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)# exit
Router(config)# interface gigabitEthernet 0/0.30
Router(config-subif)# encapsulation dot1Q 30
Router(config-subif)# ip address 192.168.30.1 255.255.255.0
Router(config-subif)# end
Router# copy running-config startup-config
```

Verify with show ip interface brief, show ip route (three connected C routes), and show interfaces trunk on the switch.

4. Task 3: PC Addressing and Test

Set each PC's IP address, subnet mask 255.255.255.0 and default gateway as in the table in section 2. Then ping the gateway first, then the other VLANs:

Ping	Expected	Note
PC1 to 192.168.10.1	Success	Own gateway
PC1 to PC2 (Foodies to Shoppers)	Success	Routed by the sub-interfaces
PC2 to PC3 (Shoppers to Streamers)	Success	Routed by the sub-interfaces
PC1 to PC3 (Foodies to Streamers)	Success	Routed by the sub-interfaces

The first ping may time out while ARP completes. Inter-VLAN traffic goes up the trunk to the router and back down the same link, which is why it is called a 'stick'.

5. Task 4: Troubleshooting Scenario

Scenario: Shoppers (PC2) cannot ping Streamers (PC3), although Foodies can still reach Shoppers.

Step 1: Isolate the problem.

- Ping from PC3 to its own gateway 192.168.30.1. It fails, so the problem is between PC3 and the router, not in the Shoppers VLAN.

- PC1 to PC2 works, which shows the trunk and the Foodies and Shoppers sub-interfaces are fine.

Step 2: Check the configuration.

```
Router# show ip interface brief
Router# show running-config | section Gi0/0.30
Switch# show vlan brief
Switch# show interfaces trunk
```

Fault found in this scenario: the sub-interface has the wrong VLAN tag.

```
interface GigabitEthernet0/0.30
 encapsulation dot1Q 300
ip address 192.168.30.1 255.255.255.0
```

The router listens for VLAN 300, but the PC3 port is in VLAN 30, so the tagged frames are never matched.

Step 3: Fix and verify.

```
Router(config)# interface gigabitEthernet 0/0.30
Router(config-subif)# encapsulation dot1Q 30
Router(config-subif)# end
```

Ping PC3 to its gateway, then PC2 to PC3. Both now succeed.

Other faults to check with the hint (VLAN assignments, sub-interfaces, trunk):

Fault	Symptom	Fix
Access port in the wrong VLAN	That PC cannot reach its gateway	switchport access vlan 30 on Fa0/4
Switch link to router is access, not trunk	No inter-VLAN traffic at all	switchport mode trunk on Fa0/1
VLAN missing from trunk allowed list	One VLAN cannot reach the router	switchport trunk allowed vlan add 30
Wrong sub-interface IP or mask	Gateway unreachable or wrong subnet	Correct the ip address command
Wrong default gateway on the PC	Local pings work, other VLANs fail	Set the gateway to the sub-interface IP
Physical interface shut down	All sub-interfaces down	no shutdown on Gi0/0
VLAN not created on the switch	Ports show in no active VLAN	Create it with the vlan command

6. Task 5: Router-on-a-Stick vs Layer 3 Switch

A Layer 3 switch routes with SVIs. Example (Packet Tracer 3560):

```
Switch(config)# ip routing
Switch(config)# interface vlan 10
```

```
Switch(config-if)# ip address 192.168.10.1 255.255.255.0
Switch(config-if)# no shutdown
```

Method	Advantages	Disadvantages
Router-on-a-stick	1. Low cost: uses a router and a single link you may already own. 2. Simple to set up and understand for small networks.	1. The single trunk link is a bandwidth bottleneck and single point of failure. 2. Software routing on the router is slower and does not scale well as VLANs and traffic grow.
Layer 3 switch	1. Fast hardware routing at wire speed, with no external link to congest. 2. Scales easily: add an SVI per VLAN and keep routing inside the switch.	1. Higher cost than a basic switch and router pair. 2. More complex to configure and fewer WAN features than a router.

7. Conclusion

Router-on-a-stick lets VLANs communicate through one trunk and sub-interfaces. Troubleshooting works best step by step: test the gateway, then check the VLAN assignment, trunk and sub-interface settings. For larger networks, a Layer 3 switch is the better choice.